

Lecture 9: Evaluating Trade-Offs

ECON7720: Ecological and Environmental Economics

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Semester 2, 2026 | Required reading: Tietenberg & Lewis, Ch. 3

Recap of Lectures 7 and 8

- GHGs differ in GWP and lifetime; CO₂ is the dominant driver of long-run warming, with impacts spanning sea level rise, ocean acidification, food security, health, and biodiversity loss.
- Tipping points, fat-tailed risks, and the carbon budget frame the urgency of action.
- Three strategies: mitigation, adaptation, and sequestration; the SCC links damages to policy.
- Carbon taxes vs. cap-and-trade: hybrid designs combine advantages; offsets extend reach but raise integrity concerns.
- International cooperation is undermined by free-riding; Barrett's model shows stable coalitions are small.

Lecture Outline

1. Benefit–cost analysis: decision rule, measuring benefits and costs, Kaldor–Hicks
2. Monetisation and discounting: WTP, VSL, present value, the discount rate debate
3. Applying BCA: sensitivity analysis, SCC, risk, irreversibility, alternatives to BCA

Benefit–Cost Analysis

The benefit–cost decision rule

The simplest rule in environmental economics

- Let B = benefits from a proposed action, C = costs.
- **Decision rule:** if $B > C$, support the action; otherwise, oppose it.
- This is the foundation of **benefit–cost analysis** (BCA), also called cost–benefit analysis (CBA).
- Governments worldwide require formal BCA for regulations with significant economic impact (see BCA frameworks slide).

But how do we measure B and C ?

- Benefits and costs must be expressed in **monetary terms** to be comparable.
- Environmental benefits (clean air, biodiversity) are often **hard to monetise**, a major challenge we will address.
- All costs should be measured as **opportunity costs**: the net benefit foregone when resources are diverted to the proposed use.

BCA frameworks and guidance documents

Government guidelines

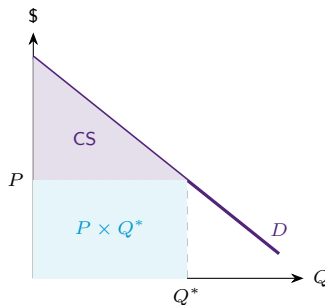
- **Australia** — Office of Impact Analysis, *Cost-Benefit Analysis Guidance Note* (2023). [\[Link\]](#)
- **United States** — OMB *Circular A-4: Regulatory Analysis* (revised November 2023). [\[Link\]](#)
- **United Kingdom** — HM Treasury, *The Green Book: Appraisal and Evaluation in Central Government* (2026 ed.; declining discount rates for >30 yr projects). [\[Link\]](#)
- **European Union** — European Commission, *Better Regulation Toolbox* (2025). [\[Link\]](#)

International organisations

- **OECD** — *Cost-Benefit Analysis and the Environment: Further Developments and Policy Use* (2018). [\[Link\]](#)

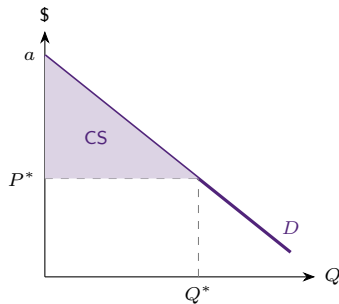
Measuring total benefits

- Total benefit = total **willingness to pay** = area under the demand curve from the origin to the quantity of interest.
- The demand curve reflects marginal willingness to pay at each unit.
- **Consumer surplus** = total willingness to pay – amount actually paid = area between demand and price.
- **Total benefit** = entire area under D (CS + expenditure $P \times Q^*$).
- **Net benefit** = total benefit – total cost.



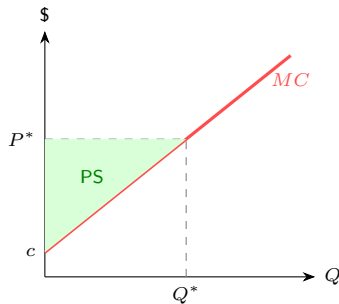
Consumer surplus explained

- **Consumer surplus (CS)** measures the benefit to buyers from purchasing at the market price.
- Each unit is valued at the buyer's **marginal willingness to pay** (the height of the demand curve).
- Because the price is below many buyers' WTP, they enjoy a **surplus** on each unit purchased.
- CS = area *between* the demand curve and the price line.
- If $P = a - bQ$, then $CS = \frac{1}{2}(a - P^*) \times Q^*$.



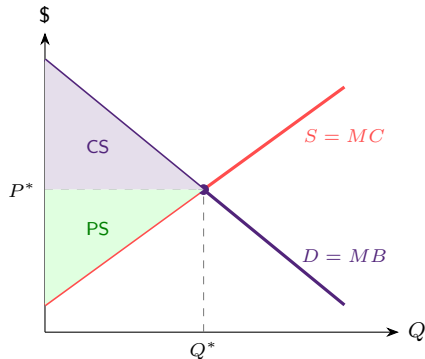
Producer surplus explained

- **Producer surplus** (PS) measures the benefit to sellers from producing at the market price.
- Each unit costs the seller the **marginal cost** (the height of the supply curve).
- Because the price exceeds many producers' MC, they enjoy a **surplus** on each unit sold.
- PS = area *between* the price line and the supply curve.
- If $MC = c + dQ$, then $PS = \frac{1}{2}(P^* - c) \times Q^*$.



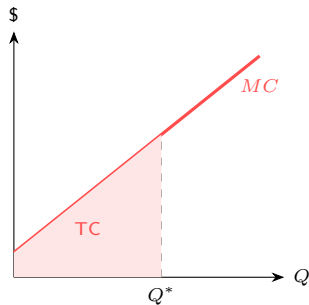
$$\text{Net benefit} = \text{CS} + \text{PS}$$

- Net benefit (economic surplus) = $\text{CS} + \text{PS}$.
- At (Q^*, P^*) , net benefit is maximised.
- Below Q^* : $MB > MC \Rightarrow$ net benefit forgone.
- Above Q^* : $MC > MB \Rightarrow$ net benefit destroyed.



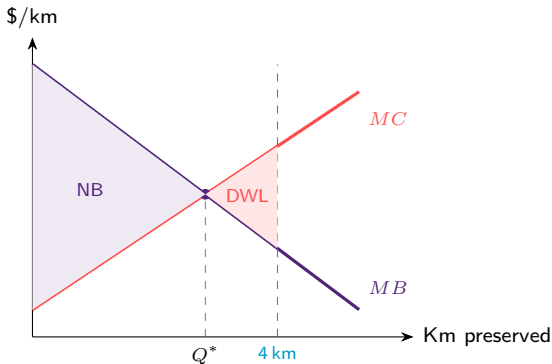
Measuring total costs

- Total cost = area under the **marginal cost** (supply) curve from the origin to Q^* .
- In competitive markets, the supply curve equals the marginal opportunity cost curve.
- **Opportunity cost**: the net benefit lost when resources are diverted from their next-best use.
- Marginal opportunity cost is the cost of producing the *last* unit.
- **Net benefit** = total benefit – total cost = CS + PS.
- An allocation that **maximises** net benefit is *efficient*.



Example: should we conserve a 4 km river stretch?

- $MB > MC$ for the first Q^* km $\Rightarrow$ net benefit is **positive** up to Q^* .
- At 4 km: $MC > MB \Rightarrow$ the 4th km costs more than it is worth.
- BCA says: preserve up to Q^* (where $MB = MC$), not necessarily all 4 km.
- The **optimal** level maximises net benefit, not the amount preserved.



Static efficiency and the First Equimarginal Principle

Static efficiency

An allocation of resources satisfies the **static efficiency** criterion if the economic surplus (net benefit) from the use of those resources is *maximised* by that allocation.

First Equimarginal Principle (the “Efficiency Equimarginal Principle”)

Net benefits are maximised when $MB = MC$.

- If $MB > MC$: net benefits can be increased by **expanding** the activity.
- If $MC > MB$: net benefits can be increased by **contracting** the activity.
- At $MB = MC$: no further reallocation can increase net benefits $\Rightarrow$ **efficient**.

Pareto optimality

Definition

An allocation is **Pareto optimal** if no other feasible allocation could benefit at least one person without any deleterious effect on some other person.

- Allocations that do *not* satisfy this condition are **suboptimal**, they can always be rearranged so that some people are made better off without harming anyone else.
- An efficient allocation (one that maximises net benefits) is Pareto optimal.
- Pareto optimality is a **necessary condition** for efficiency, but it is *silent on equity*, a Pareto optimal allocation can still be very unequal.

Practical implication

BCA identifies **potential** Pareto improvements: if $B > C$, winners could in principle compensate losers and still be better off. Whether compensation actually occurs is a **distributional** question.

From efficiency to optimality

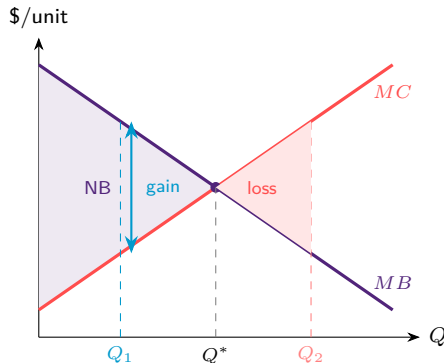
- **Efficiency criterion:** $MB = MC$ maximises net benefits, an abstract principle.
- **Optimal allocation:** the specific level of the activity that satisfies $MB = MC$ in practice, requires measuring MB and MC , choosing a discount rate, defining the accounting stance, and valuing non-market effects.
- BCA is the **operational tool** that translates the efficiency criterion into a concrete policy recommendation.

The policy logic

1. Use BCA to identify the **optimal** outcome.
2. Measure the **divergence** between the actual and optimal allocations (e.g. overfishing, excessive emissions, under-conservation).
3. Design **policy instruments** to close the gap (taxes, cap-and-trade, subsidies, protected areas).

MB = MC maximises net benefit

- At $Q_1 < Q^*$: $MB > MC \Rightarrow$ each additional unit adds more benefit than cost. **Expand**.
- At $Q_2 > Q^*$: $MC > MB \Rightarrow$ each additional unit destroys net benefit. **Contract**.
- Only at Q^* where $MB = MC$ is there no gain from reallocation: net benefit is **maximised**.



Opportunity cost in environmental contexts

Opportunity cost = the net benefit foregone from the next-best alternative use of a resource.

- **Forest preservation vs. logging:**

- Opportunity cost of preservation = forgone timber revenue, mining access, and agricultural conversion.
- Opportunity cost of logging = lost ecosystem services (carbon storage, water filtration, biodiversity, recreation).

- **Wetland conservation vs. development:**

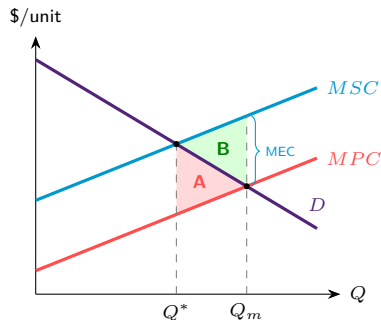
- Opportunity cost of conservation = forgone residential or commercial development revenue.
- Opportunity cost of development = lost flood control, water purification, and habitat services.

- **Water allocation:** water used for irrigation cannot simultaneously support river ecosystems $\Rightarrow$ the Murray–Darling conflict.

BCA requires that *all* costs be measured as opportunity costs, not just accounting costs.

The Kaldor–Hicks criterion

- A **Pareto improvement** makes at least one person better off and no one worse off, extremely rare in practice.
- A **potential Pareto improvement** (Kaldor–Hicks): the winners could *in principle* compensate the losers and still be better off (i.e. $B > C$).
- Compensation is **hypothetical**, BCA does not require that it actually occurs.
- This is the welfare criterion that underpins BCA: a policy passes if **total gains exceed total losses**.



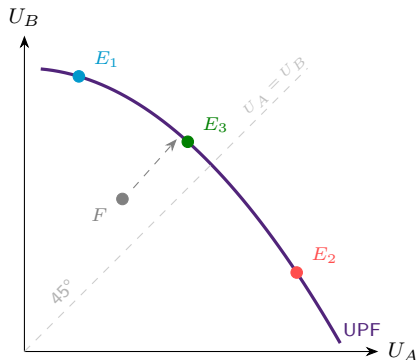
A = private surplus lost; B = DWL eliminated.
Social gain = $A + B$; private loss = A .

$B > 0 \Rightarrow$ gains exceed losses.

Efficiency vs. equity

- Kaldor–Hicks addresses **efficiency**; it is silent on the **distribution** of gains and losses.
- E_1 , E_2 , E_3 are all **Pareto optimal**, yet they imply very different distributions.
- F is **Pareto suboptimal** (unrealised gains from trade exist), yet may rank higher on equity grounds.

Efficiency determines whether an allocation lies *on* the frontier; it does **not** determine *where* on the frontier.



UPF = Utility Possibility Frontier: the set of all Pareto optimal allocations between two agents.

Distribution of benefits and costs

BCA tells us whether $B > C$, but not *who* gains and who loses

- **Economic impact analysis:** a broad characterisation of the gains and losses from a given policy.
- **Equity analysis:** examines impacts on disadvantaged groups or sub-populations.

Distributional implications

- **Proportional:** same proportional impact regardless of income level.
- **Regressive:** higher proportional net benefits to high-income vs. low-income groups (or higher proportional costs on low-income).
- **Progressive:** higher proportional net benefits to low-income groups.
- A policy with $B > C$ but **regressive** impacts may be politically unacceptable without compensating transfers.
- Recall: carbon pricing is efficient but often regressive $\Rightarrow$ revenue recycling matters (Lecture 7).

Market failure as divergence from optimality

Connecting back to Lectures 1–2

- In a perfectly competitive market with no externalities, the equilibrium is **Pareto optimal** (First Welfare Theorem).
- **Market failures** create a wedge between the market outcome and the social optimum:
 1. **Externalities**: costs/benefits not reflected in prices (pollution, congestion).
 2. **Public goods**: non-rival and non-excludable $\Rightarrow$ under-provision (clean air, biodiversity).
 3. **Common-pool resources**: rival but non-excludable $\Rightarrow$ over-exploitation (fisheries, aquifers).
 4. **Information asymmetries**: adverse selection, moral hazard.
 5. **Market power**: monopoly pricing.
- BCA helps quantify the **gap** between actual and optimal outcomes, guiding policy design.

BCA as a normative vs. positive tool

Two perspectives

- **Positive analysis**: describes what *is*, the current state and likely consequences of a policy.
- **Normative analysis**: prescribes what *ought* to be, whether a policy *should* be adopted.
- BCA is fundamentally a **normative** tool: it recommends actions based on the criterion that net benefits should be maximised.

BCA is not value-free

Every BCA embeds value judgements, the **welfare criterion** (WTP favours the wealthy), the **discount rate** (intergenerational equity), and the **accounting stance** (whose welfare counts). Its strength is that it makes these judgements **explicit and transparent**.

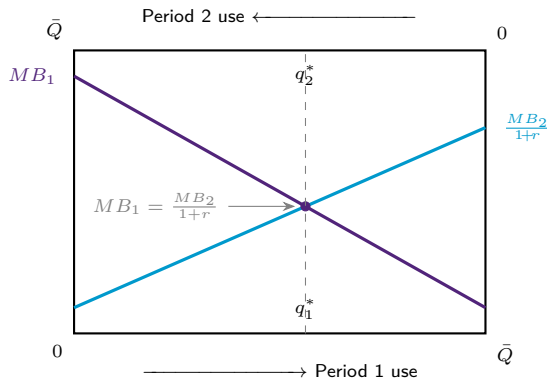
From static to dynamic efficiency

Static efficiency is not enough

$MB = MC$ maximises net benefit at a **single point in time**, but environmental decisions, resource depletion, pollution accumulation, climate damages, persist across periods.

Dynamic efficiency

Maximise the **present value** of net benefits across all periods. A higher r favours present use; for long-lived resources, r can **reverse** the recommendation.



A fixed resource $\bar{Q}$ is allocated between two periods. The dynamically efficient split equates MB_1 with the **discounted** MB_2 . A higher r tilts allocation toward the present.

Present value: the basic formula

One-time payment

The present value of $\$B$ received n years from now:

$$PV = \frac{B_n}{(1+r)^n}$$

where r is the **discount rate**.

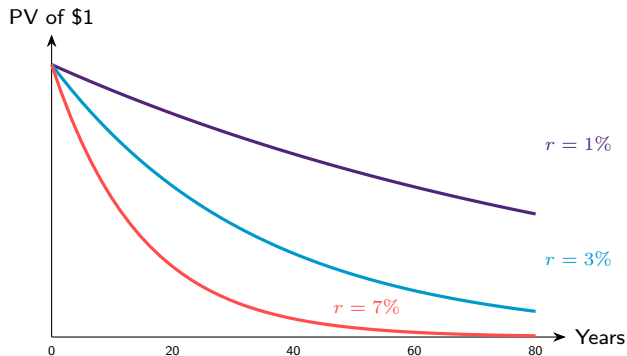
Stream of payments

The present value of a stream $\{B_0, B_1, \dots, B_n\}$:

$$PV = \sum_{i=0}^n \frac{B_i}{(1+r)^i}$$

- **Discounting** is the process of calculating present value.
- Higher $r \Rightarrow$ future amounts are worth **less** today.
- Lower $r \Rightarrow$ future amounts retain **more** of their value.

Visualising the discount rate



- At $r = 7\%$, \$1 received 40 years from now is worth only \$0.07 today.
- At $r = 1\%$, the same dollar is worth \$0.67.
- For climate change (damages 50–200+ years out), the discount rate is the single most influential parameter.

Continuous vs. discrete discounting

Discrete discounting (standard)

$$PV = \frac{B_n}{(1+r)^n}$$

Assumes compounding once per year.

Continuous discounting

$$PV = B_n \cdot e^{-rn}$$

Assumes instantaneous compounding.

- Continuous discounting is used in some theoretical models (e.g. Ramsey growth model) and in financial engineering.
- For small r and moderate n , the difference is **small**:
 - Discrete: $1/(1.05)^{10} = 0.6139$
 - Continuous: $e^{-0.05 \times 10} = 0.6065$
- For large n (climate change), the difference grows.
- In environmental BCA, **discrete discounting** is standard practice.

Why do we discount, and which rate?

Three reasons to discount

Reason	Logic
Time preference	Prefer consumption sooner
Opportunity cost	\$1 invested grows to $\$(1+r)$
Uncertainty	Future may not arrive as expected

- Two components: **risk-free rate** (government bonds) + **risk premium**.
- The most **consequential assumption** in any BCA.

Social vs. private rates

- **Private**: individual time preference + firm's opportunity cost of capital.
- **Social**: society's collective time preference + social risk premium.
- Social rate is typically **lower** $\Rightarrow$ society should invest more in long-term environmental protection than private agents would.
- They diverge when society values **intergenerational equity** more than markets do.

Declining discount rates

Weitzman (2001): uncertainty about the discount rate

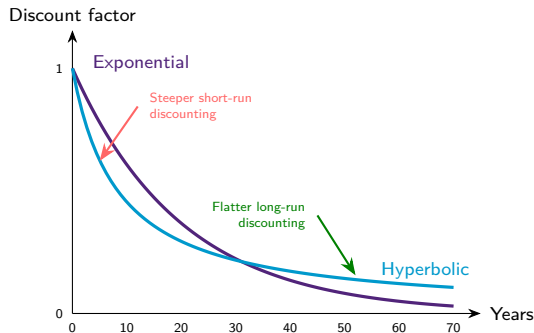
- If we are **uncertain** about the correct constant discount rate, the *effective* discount rate applied to distant-future outcomes should **decline over time**.
- Intuition: low rates dominate in the long run because they shrink the discount factor least.
- Weitzman surveyed 2,160 economists $\Rightarrow$ average recommended rate $\approx 4\%$, but wide disagreement.

Policy adoption

- **UK HM Treasury** (Green Book): uses a declining schedule:
 - Years 0–30: 3.5%
 - Years 31–75: 3.0%
 - Years 76–125: 2.5%
 - Years 126–200: 2.0%
 - Years 200+: 1.5%
- France also uses declining rates. Australia does **not** (yet).

Hyperbolic discounting: a behavioural perspective

- **Exponential**: constant discount rate $\Rightarrow$ time-consistent preferences.
- **Hyperbolic**: steep discounting in the short run, flatter in the long run $\Rightarrow$ time-inconsistent preferences.
- Plans made today are reversed when the future arrives (Laibson, 1997).
- Environmental policy implication: governments commit to ambitious long-run targets but **delay** costly near-term action.



Applying Benefit–Cost Analysis and Alternatives

Issues in benefit estimation: overview

Five key challenges

1. **Primary vs. secondary effects:** which effects to count.
2. **Accounting stance:** the geographic scale at which benefits are measured (local, national, global).
3. **Aggregation:** how to sum benefits across individuals and groups.
4. **With-and-without principle:** only count benefits that would *not* have accrued without the project.
5. **Tangible vs. intangible benefits:** tangible benefits can be assigned a monetary value; intangible benefits (aesthetic, spiritual, existence value) cannot easily be monetised.

Primary vs. secondary effects

What should be counted?

- **Primary effects:** direct consequences of the policy (e.g. reduced pollution $\Rightarrow$ health improvements).
- **Secondary effects:** indirect consequences (e.g. new jobs created by a conservation project).
- **Rule:** secondary employment benefits should be counted *only* in areas with high unemployment or underemployed skills.
- If the project simply results in a **rearrangement** of productively employed resources, the employment effect is a *transfer*, not a net benefit.

Common mistake

Students (and analysts) often count secondary employment on the benefit side even when labour is simply redirected from one productive use to another. This **double counts** and inflates the BCA.

Accounting stance and the with-and-without principle

Accounting stance

- The **geographic scope** at which benefits and costs are measured matters:
 - A local dam project may have positive local B but negative national B (e.g. if downstream effects are counted).
 - Climate policy: benefits are global; costs are often national.
- The choice of stance can **reverse** the BCA result.

The with-and-without principle

- Only count benefits that would result **from the project**, ignoring those that would have accrued anyway.
- Example: if a wetland would have attracted tourists regardless of a new boardwalk, the boardwalk's benefit is only the *incremental* tourists, not the total.
- Violating this principle is a common source of **overestimation** in BCA.

Limitations of WTP as a measure of value

Why willingness to pay is imperfect

- **Income effects:** WTP depends on ability to pay.
 - A wealthy person's WTP for clean air $>$ a poor person's WTP, not because they value it more, but because they *can afford to pay more*.
 - This means BCA, unweighted, gives more voice to the rich.
- **Hypothetical bias:** in stated-preference surveys, people may overstate or understate their true WTP because no real payment is required.
- **Scope insensitivity:** stated WTP often fails to scale with the magnitude of the good.
- **Status quo bias:** WTP to gain a good $<$ willingness to accept (WTA) to give it up, the “endowment effect.”

We will explore these issues further in [Lecture 10](#) when we cover non-market valuation methods.

Tangible vs. intangible benefits and non-market goods

The monetisation problem

- **Tangible:** can be assigned a monetary value (avoided medical costs, crop yields, timber revenue).
- **Intangible:** cannot easily be monetised (aesthetic beauty, cultural heritage, existence value, climate stability).
- E.g. infrastructure BCA has market prices for most benefits; environmental BCA often does **not**.

Solutions (Lecture 10)

- **Revealed preference:** infer WTP from observed behaviour (hedonic pricing, travel cost).
- **Stated preference:** ask WTP directly (contingent valuation, choice experiments).
- Revealed preference is more credible but limited to **use values**; stated preference can capture **non-use values** but is vulnerable to biases.

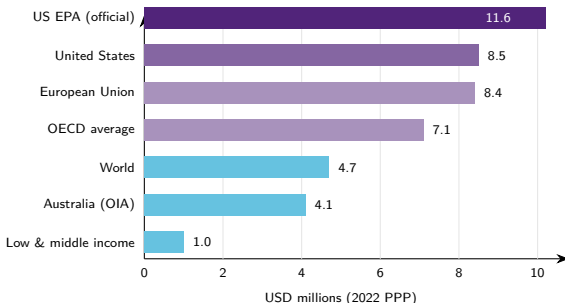
Example: the value of a statistical life (VSL)

Definition

VSL = aggregate WTP for a small reduction in **mortality risk**.

- 10,000 people $\times$ \$700 for a 1-in-10,000 risk reduction = one statistical life = \$7M.
- Does **not** price an identified life; it values *marginal risk changes*.
- Estimated via **hedonic wages** or **stated preference surveys**.
- Dominates air-quality BCA ($\sim 85\text{--}90\%$ of benefits); the choice of VSL can **determine** whether a regulation passes $B > C$.

VSL estimates (USD millions, 2022 PPP)



Source: OECD (2025), *Mortality Risk Valuation in Policy Assessment*; US EPA; Australia OIA. [OECD podcast: What is the VSL?](#)

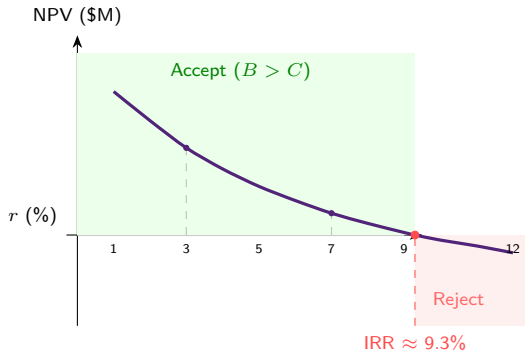
Approaches to cost estimation

1. **The survey approach:** ask polluters/firms about their control costs.
 - Advantage: uses actual firm-level information.
 - Disadvantage: firms may **overstate** costs strategically (to argue against regulation).
 2. **The engineering approach:** use engineering information to estimate available technologies and their costs.
 - Advantage: independent, based on technical analysis.
 - Disadvantage: may not reflect real-world implementation (“typical” firm $\neq$ actual firm).
 3. **The combined approach:** use both survey and engineering data.
 - Most common in practice.
 - Cross-validates estimates from different sources.
-
- Estimating costs is generally **more straightforward** than estimating benefits, but still subject to uncertainty about future technologies and compliance behaviour.

Sensitivity analysis

When estimates are uncertain, test their robustness

- BCA results depend on **assumptions**: discount rate, benefit magnitudes, cost projections, time horizon.
- **Sensitivity analysis**: systematically vary key parameters to see how the result changes.
- If $B > C$ is **robust** to reasonable parameter ranges, confidence is high.
- If the conclusion **flips**, the decision is ambiguous $\Rightarrow$ further investigation or precautionary reasoning.



Example: \$50M cost today, \$5M/yr benefits for 30 years. At $r = 3\%$: NPV = +\$48M (accept). At $r = 7\%$: NPV = +\$12M (accept). At $r = 10\%$: NPV = -\$3M (reject). The discount rate is **decisive**.

Application: the social cost of carbon

- SCC = present value of marginal damages from one additional tonne of CO_2 emitted.
- Control costs are straightforward;
benefits = avoided damages = the SCC .

Evolution of estimates

Year	SCC	Discount rate
2010 (US govt)	\$22/t	3%
2016 (IWG update)	\$37/t	3%
2022 (Rennert et al.)	\$185/t	2%
2023 (EPA proposed)	\$190/t	2%

Sources: IWG (2016), *Technical Update of the Social Cost of CO_2* ; Rennert et al. (2022), *Nature* 610, 687–692; US EPA (2023).

SCC by discount rate (\$/tonne, 2020\$)

Year	5%	3%	2.5%	3% (95th)
2020	\$14	\$51	\$76	\$152
2030	\$19	\$62	\$89	\$187
2040	\$23	\$73	\$103	\$221
2050	\$26	\$85	\$116	\$260

95th pct. = 95th percentile of the damage distribution, capturing high-damage, low-probability (fat-tail) scenarios.

- The **discount rate** and **risk treatment** together determine whether an optimal carbon tax is \$25 or \$200.
- The 2022/2023 revision to approximately \$190 reflects lower discount rates and improved damage functions.

The treatment of risk: dominant policies

When decisions are easy

- A **dominant policy** is one that confers the highest net benefits *in every possible outcome*.
- If such a policy exists, the decision is straightforward: choose it.
- But dominant policies are rare in environmental settings— usually different outcomes favour different policies.

Expected value approach

- The **expected present value** of net benefits:

$$E[NPV_j] = \sum_i \pi_i \cdot PV(NB_{ij})$$

where π_i is the probability of outcome i and NB_{ij} is the net benefit of policy j in outcome i .

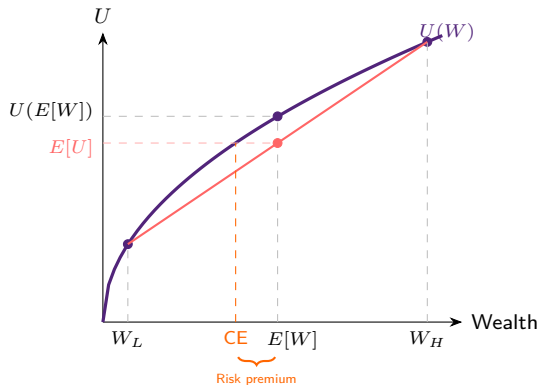
- Choose the policy with the **highest expected NPV**.

Risk neutrality, aversion, and irreversibility

- **Risk neutral**: only expected value matters
 $\Rightarrow$ standard BCA assumption.
- **Risk averse**: a certain outcome is preferred to a gamble with the same expected value (concave U).
- **Risk loving**: gambles preferred (convex U).

Irreversibility

- If a species goes extinct or a tipping point is crossed, the decision **cannot be undone**.
- The **option value** of waiting may justify delaying irreversible actions, even if expected NPV is positive.



Risk premium = $E[W] - CE$: the amount a risk-averse agent would pay to avoid the gamble.

The precautionary principle

Rio Declaration (1992), Principle 15

“Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures.”

- Applied when probabilities are **unknown** (“deep uncertainty”), damages are potentially **catastrophic and irreversible**, and standard expected-value BCA is unreliable.
- It shifts the **burden of proof**: from “prove harm before regulating” to “prove safety before proceeding.”
- **Criticism**: if interpreted too broadly, it can block *any* action, including beneficial innovation.
- **Defence**: it is a **decision rule for uncertainty**, not a ban on risk-taking, it says act preventively when consequences are severe.
- Examples: **EU Directive 2001/18** (GMO releases require prior safety assessment); **REACH** (chemicals must be proven safe before market placement).

Option value and quasi-option value

The value of keeping options open

- **Option value**: the value of maintaining the *option* to use a resource in the future, even if you are not sure you will use it.
 - Analogous to a financial option: you pay a premium for the right (not the obligation) to act later.
- **Quasi-option value** (Arrow & Fisher, 1974): the expected value of *information* gained by delaying an irreversible decision.
 - If you destroy a rainforest today, you lose the ability to learn its value and act on that knowledge.
 - Quasi-option value is always **non-negative**: more information cannot make you worse off.
- Implication: standard BCA without option/quasi-option value may **over-recommend** irreversible development.
- This strengthens the case for **conservation** when future values are uncertain.

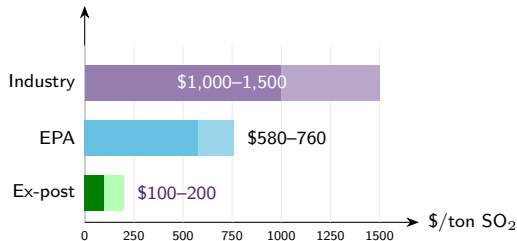
Ex-ante vs. ex-post BCA

Why predictions often differ from reality

- **Ex-ante BCA**: conducted *before* the policy is implemented, based on projections.
- **Ex-post BCA**: conducted *after* implementation, using actual data.
- Systematic patterns in the gap:
 - **Costs are often overestimated** ex ante: firms find cheaper compliance methods than predicted (innovation, process changes).
 - **Benefits are often underestimated** ex ante: co-benefits (health, productivity) are discovered later.
 - **Optimism bias** in *non-regulatory* projects (Flyvbjerg et al., 2003): infrastructure costs are systematically underestimated (average overrun: 28% for roads, 45% for rail).
- The direction of bias depends on whether costs or benefits are being projected, and by whom.

Case study: the US SO₂ Acid Rain Program

- Ex-ante industry forecast:
\$1,000–1,500/ton.
- Ex-post allowance prices:
\$100–200/ton (5–10× lower).
- Cap-and-trade enabled **fuel switching**, **innovation**, and **flexibility**, costs fell far below mandated-technology estimates.
- Benefits (avoided PM_{2.5} mortality):
B/C > 30:1.



Ex-ante cost estimates vs. actual allowance prices. Source: Schmalensee & Stavins (2013), *Journal of Economic Perspectives* 27(1), 103–122.

Double counting in BCA: common pitfalls

Errors that inflate benefit or cost estimates

1. **Counting both WTP and cost savings:** if improved air quality raises property values (hedonic) *and* you separately count reduced medical costs, you may double-count because property values already capitalise health benefits.
2. **Counting producer and consumer surplus twice:** counting increased tourism revenue (producer surplus) *and* tourist WTP (consumer surplus) when they overlap.
3. **Counting secondary employment as a benefit:** if workers are drawn from other productive employment, the net employment benefit is zero (it is a transfer).
4. **Mixing stocks and flows:** counting a one-time benefit as if it recurs annually, or vice versa.
5. **Ignoring displacement:** a new national park may increase visits to one area but reduce visits to substitutes.

Rule of thumb: always ask “Is this benefit already captured elsewhere in the analysis?”

Arrow et al. (1996): principles for BCA

Guidance from economists

1. BCA is a useful framework for **organising information**; it should not be the sole basis for decision-making.
2. Benefits and costs should be **quantified** wherever possible, but unquantifiable effects must also be considered.
3. BCA should be **transparent**: assumptions, data sources, and sensitivity analysis must be documented.
4. **Distributional effects** should be presented alongside aggregate results.
5. The **discount rate** should reflect social opportunity cost, with sensitivity analysis.
6. Compare the proposed action against the **best alternative**, not just “do nothing.”
7. **Ex-post evaluation** should be conducted to improve future analysis.
8. “BCA can greatly improve the process and, hence, the outcome of policy analysis” if properly done.

A critical appraisal of BCA

Strengths

- Provides a **systematic, transparent** framework for comparing options.
- Forces analysts to identify and quantify trade-offs explicitly.
- Ensures **consistency** across policy evaluations.

Limitations

- Monetising intangible environmental benefits is **difficult and controversial**.
- Results are highly sensitive to the **discount rate** and other assumptions.
- May systematically undervalue **long-term, irreversible** environmental damages.
- Distributional impacts are not captured by aggregate $B > C$.
- Risk of “garbage in, garbage out”: unreliable estimates limit the value of BCA.

Arrow et al. (1996): “BCA can provide an exceptionally useful framework for consistently organising disparate information [...] it can greatly improve the process and, hence, the outcome of policy analysis.”

Cost-effectiveness analysis

When benefits cannot be reliably measured

- BCA requires monetised benefits, but sometimes benefits are **unmeasurable** or too controversial to monetise.
- **Cost-effectiveness analysis** (CEA) offers an alternative:
 1. Set a policy **target** on some basis other than BCA (e.g. a health-based pollution threshold, a temperature limit, a species survival probability).
 2. Find the **least-cost means** of achieving that target.
- CEA does *not* tell us whether the target itself is efficient, only how to achieve it at minimum cost.

Relationship to efficiency

- All efficient policies are cost-effective, but **not all cost-effective policies are efficient** (the target may not be optimal).

The Cost-Effectiveness Equimarginal Principle

The least-cost means of achieving an environmental target is reached when the **marginal costs** of all possible means of achievement are equal.

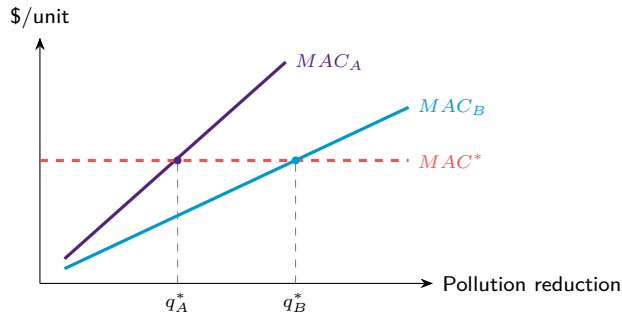
- **Intuition:** if $MC_A > MC_B$, shift effort from A to $B \Rightarrow$ same total reduction at lower cost.
- Once $MC_A = MC_B = \dots = MC_n$, no further cost savings are possible.
- This is the same logic as the equimarginal principle in pollution control (Lectures 3–4): equalise MAC across all sources.

Application

In pollution control, CEA can:

1. Find the least-cost means of meeting a given standard.
2. Estimate how much costs would **increase** from this minimum if non-cost-effective policies are used.
3. Assess how compliance costs change under more or less stringent standards.

Cost-effectiveness: a graphical example



- Target total reduction: $q_A^* + q_B^*$. Cost-effectiveness requires $MAC_A = MAC_B = MAC^*$.
- Source B reduces more (lower MAC); source A reduces less $\Rightarrow$ same total at **minimum cost**.
- A uniform standard forcing equal reductions would cost **more**.

Impact analysis

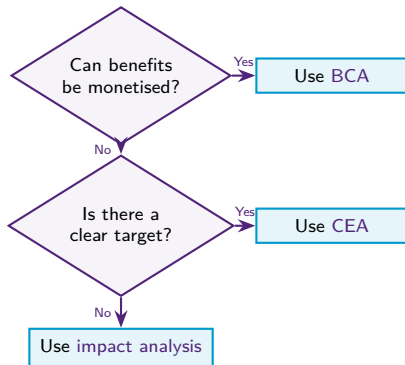
When neither benefits nor costs can be reliably measured

- An **impact analysis** attempts to quantify the *consequences* of various actions without converting them into a single monetary measure.
- No optimisation; no attempt to compare benefits and costs directly.
- Places a large amount of relatively **undigested information** at the disposal of the policy-maker.
- The policy-maker must then make a **judgement call** about whether the trade-offs are acceptable.

Environmental Impact Assessment (EIA)

- EIA originated as pure impact analysis but today often includes BCA or CEA components.
- Required in most jurisdictions for major development projects (e.g. mining, infrastructure).
- Provides structured information on environmental, social, and economic impacts.

Choosing the right tool: BCA, CEA, or impact analysis



	BCA	CEA	Impact
Requires	Monetised B & C	Monetised C	Neither
Tells us	Is $B > C$?	Cheapest way	What happens?
Optimises?	Yes	Conditional	No
Main use	Go/no-go	B hard to measure	B & C uncertain

- CEA outcome measures: cost per QALY, cost per tonne CO₂ abated, cost per hectare protected.
- In practice, most evaluations use a **combination** of all three.

Appendix

References (1/2)

- Arrow, K.J. & Fisher, A.C. (1974), "Environmental Preservation, Uncertainty, and Irreversibility," *Quarterly Journal of Economics* 88(2), 312–319.
- Arrow, K.J. & Lind, R.C. (1970), "Uncertainty and the Evaluation of Public Investment Decisions," *American Economic Review* 60(3), 364–378.
- Arrow, K.J., Cropper, M.L., Eads, G.C., Hahn, R.W., Lave, L.B., Noll, R.G., Portney, P.R., Russell, M., Schmalensee, R., Smith, V.K. & Stavins, R.N. (1996), "Is There a Role for Benefit–Cost Analysis in Environmental, Health, and Safety Regulation?" *Science* 272(5259), 221–222.
- Flyvbjerg, B., Bruzelius, N. & Rothengatter, W. (2003), *Megaprojects and Risk: An Anatomy of Ambition*, Cambridge University Press.
- IWG (2016), *Technical Update of the Social Cost of Carbon for Regulatory Impact Analysis*, US Government Interagency Working Group on Social Cost of Greenhouse Gases.
- Kahneman, D. (1986), "Comments on the Contingent Valuation Method," in *Valuing Environmental Goods*, ed. R.G. Cummings, D.S. Brookshire & W.D. Schulze, Rowman & Allanheld, 185–194.

References (2/2)

- Laibson, D. (1997), "Golden Eggs and Hyperbolic Discounting," *Quarterly Journal of Economics* 112(2), 443–478.
- Nordhaus, W.D. (2007), "A Review of the Stern Review on the Economics of Climate Change," *Journal of Economic Literature* 45(3), 686–702.
- OECD (2025), *Mortality Risk Valuation in Policy Assessment: A Global Meta-Analysis of Value of Statistical Life Studies*, OECD Publishing, Paris.
- Rennert, K., Errickson, F., Prest, B.C. et al. (2022), "Comprehensive Evidence Implies a Higher Social Cost of CO₂," *Nature* 610, 687–692.
- Schmalensee, R. & Stavins, R.N. (2013), "The SO₂ Allowance Trading System: The Ironic History of a Grand Policy Experiment," *Journal of Economic Perspectives* 27(1), 103–122.
- Stern, N. (2006), *The Economics of Climate Change: The Stern Review*, Cambridge University Press.
- Weitzman, M.L. (2001), "Gamma Discounting," *American Economic Review* 91(1), 260–271.